INCH-POUND

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SUPERSEDING
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FEDERAL SPECIFICATION

MILLING, GRINDING, DRILLING, AND SLOTTING ATTACHMENT

The General Services Administration has authorized the use of this federal specification by all federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers an electric motor driven attachment for milling, grinding, drilling, slotting, boring, gear cutting, thread milling, and keyseating operations. The attachment is adaptable for mounting on a lathe, planer, or boring mill. This attachment may also be used as a portable self-contained unit suitable for bench mounting when provided with a feed table.

1.2 Classification. The attachments shall be of the following sizes (see 6.2):

Size 1 - see table I
Size 2 - see table I

2. APPLICABLE DOCUMENTS

2.1 Government publications. The issues of the following documents, in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Standards

FED-STD-H28	- Screw-Thread Standards for Federal Services.
FED-STD-376	- Preferred Metric Units for General Use by the Federal Government.

Beneficial comments, recommendations, additions, deletions clarifications, etc. and any data that may improve this document should be sent to: STDZNMGT@dla.mil or Defense Supply Center Richmond, ATTN: DSCR-VEB, 8000 Jefferson Davis Highway, Richmond, VA 23297-5616.

AMSC N/A FSC 3460

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Military Standards

MIL-STD-130	- Identification Marking of U.S. Military Property.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1916	- DoD Preferred Methods for Acceptance of Product.

(Copies of military standards required by contractors in connection with specific procurement functions are obtained from the Standardization Document Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094. Electronic copies may be obtained from http://assist.daps.dla.mil/quicksearch/.)

Code of Federal Regulations (CFR)

29 CFR 1910.211 - 219, Subpart O - Machinery and Machine Guiding.

(The Code of Federal Regulations (CFR) is for sale on a subscription basis from the Superintendent of Documents, P.O. Box 371954, Pittsburgh, PA 15250-7954. When indicated, reprints of certain regulations may be obtained from the federal agency responsible for issuing them. Electronic copies may be obtained from http://www.access.gpo.gov/.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on the date of invitation for bids or request for proposal shall apply.

American Gear Manufacturers Association (AGMA)

AGMA 904	- Metric Usage (DoD adopted).
AGMA 2000	- Gear Classification and Inspection Handbook - Tolerances and Measuring Methods for Unassembled Spur and Helical Gears (Including Metric Equivalents) (DoD adopted).
AGMA 2009	- Bevel Gear Classification Tolerances and Measuring Methods.
AGMA 2011	- Cylindrical Wormgearing Tolerance and Inspection Methods.

(Private sector and civil agencies may purchase copies of these voluntary standards from American Gear Manufacturers Association, 500 Montgomery Street, Suite 350, Alexandria, VA 22314-1560. Electronic copies may be obtained from http://www.agma.org/.)

American National Standards Institute (ANSI)

ANSI B7.1	- Safety Requirements for the Use, Care and Protection of Abrasive Wheels (DoD adopted).

(Private sector and civil agencies may purchase copies of these voluntary standards from the American National Standards Institute, 25 West 43rd Street, 4th Floor, New York, NY 10036. Electronic copies may be obtained from http://web.ansi.org/.)

American Society of Mechanical Engineers (ASME)

ASME B5.10	- Machine Tapers (Self Holding and Steep Taper Series) (DoD adopted).
ASME B17.1	- Keys and Keyseats (DoD adopted).
ASME B94.19	- Milling Cutters and End Mills (DoD adopted).

(Private sector and civil agencies may purchase copies of these voluntary standards from the American Society of Mechanical Engineers, Three Park Avenue, New York, NY 10016-5990. Electronic copies may be obtained from http://www.asme.org/.)

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American Welding Society (AWS)

AWS A5.1	- Specification for Carbon Steel Electrodes for Shielded Metal Arc Welding (DoD adopted).
AWS A5.18	- Specification for Carbon Steel Filler Metals for Gas Shielded Arc Welding (DoD adopted).
AWS A5.20	- Carbon Steel Electrodes for Flux Cored Arc Welding (DoD adopted).

(Private sector and civil agencies may purchase copies of these voluntary standards from the American Welding Society, 550 N.W. LeJeune Road, Miami, FL 33126. Electronic copies may be obtained from http://www.aws.org/.)

National Electrical Manufacturers Association (NEMA)

NEMA ICS 1	- General Standards for Industrial Control and Systems (DoD adopted).
NEMA MG 1	- Motors and Generators (DoD adopted).

(Private sector and civil agencies may purchase copies of these voluntary standards from the National Electrical Manufacturers Association, 1300 North 17th Street, Suite 1847, Rosslyn, VA 22209. Electronic copies may be obtained from http://www.nema.org/.)

National Fire Protection Association (NFPA)

NFPA 70	- National Electrical Code (DoD adopted).
NFPA 79	- Electrical Standard for Metalworking Machine Tools and Plastic Machinery (DoD adopted).

(Private sector and civil agencies may purchase copies of these voluntary standards from the National Fire Protection Association, 1 Batterymarch Park, P.O. Box 9101, Quincy, MA 02269-9101. Electronic copies may be obtained from http://www.nfpa.org/.)

Rubber Manufacturers Association (RMA)

RMA IP 25	- Variable Speed V-Belts.

(Private sector and civil agencies may purchase copies of these voluntary standards from the Rubber Manufacturers Association, 1400 K Street NW, Suite 900, Washington, DC 20005. Electronic copies may be obtained from http://rma.org/.)

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3. REQUIREMENTS

3.1 First article. When specified (see 6.2), one or more complete attachments shall be subjected to first article testing inspection in accordance with paragraph 4.3. The attachments may be either a standard commercial production item, pre-production item, or initial production attachment that conforms to the requirements of this document.

3.2 Design. The attachments shall be new and of the manufacturer's latest design and shall meet the requirements of this specification. The attachments shall be powered and capable of multi-positional, vertical to horizontal, spindle axis adjustments for machine operations and ready to mount on a lathe, milling machine, planer, or boring mill. The attachments shall be portable and self contained when mounted upon a compound slide. When mounted on a lathe, the attachments shall be adaptable for mounting on the compound rest or cross slide. The design shall be such that parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, or repair without modification, distortion, or damage to the attachment or any attachment part or component.

3.2.1 Measurement system. The U.S. Customary System of Units (US) or the International System of Units (SI) shall be used in the design and construction of the attachments. In this specification, all measurements, dimensions, sizes, and capacities are given in the US system. These measurements may be converted to the SI system by using the conversion factors and methods specified in FED-STD-376.

3.2.2 Measuring and indicating devices. When specified (see 6.2), measuring and indicating devices shall be graduated in both the US and SI systems of measurement. When a dual US and SI system is furnished, feed dials shall have independent zero adjustments and shall be calibrated so that the last dial graduation progresses into, and is continuous with, the first dial graduation as the dial is rotated through the zero position. When only one system of graduation is acceptable, the particular graduation required shall be as specified (see 6.2). Regardless of the measurement system used, all measuring and indicating devices shall be graduated in the same system.

3.2.3 Threads. Threads shall conform to FED-STD-H28 and the applicable detailed standard section referenced therein.

3.2.4 Gears. All gears and pinion designs selected for use in the attachment or its components shall be manufactured to either US standard inch diametral pitch in accordance with AGMA 2000, AGMA 2009, and AGMA 2011 or SI module system in accordance with AGMA 904.

3.2.5 Safety devices. Safety devices shall be provided for parts that present safety hazards to personnel. Safety devices shall not interfere with the operation of the attachment or with the operational performance.

3.2.5.1 Occupational safety standards. The product shall comply with Occupational Safety and Health Administration (OSHA) standards set forth in 29 CFR 1910.211-219, Subpart O that are applicable to this type of equipment.

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3.3 Material. Materials not specifically designated herein or in the acquisition order shall be of a quality commensurate with commercial practice within the machine tool industry. The materials shall be suitable for the intended purpose in the design of the end item and shall meet all requirements specified herein. Materials shall be free from defects that would adversely affect the performance or maintainability of the individual components or the overall assembly. When dissimilar metals are used in contact with each other, suitable protection against galvanic corrosion shall be applied in accordance with MIL-STD-889.

3.3.1 Recycled, recovered, or environmentally preferable materials. Unless otherwise specified herein, all materials covered by this specification shall be new. However, recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements and promotes economically advantageous life cycle costs.

3.4 Construction. Parts used in fabrication of the attachments shall be clean and free of defects. External surfaces shall be smooth and free of sharp edges. Screws, pins, bolts, and similar fasteners shall be installed with means to prevent loss of tightness and adjustment. Parts subject to adjustment or removal shall not be swaged, peened, staked, or otherwise deformed.

3.4.1 Castings and forgings. Castings shall be free from visible blowholes, porosity, hard spots, shrinkage defects, cracks, or other defects. Forgings shall be free from visible scale, inclusions, cold shuts, mismatching, sharp edges, or other defects. Castings and forgings shall be free from any properties adversely affecting their strength, durability, or suitability. Reclamation or repair of major castings and forgings is permitted if the repair is not on a critical surface.

3.4.2 Welding. Welding of carbon steel base materials shall be accomplished using filler metals listed in AWS A5.1, AWS A5.18, AWS A5.20, or equivalent. These operations shall not be employed as repair measures for defective parts.

3.4.3 Electrical connections. Connections of conductors and terminal parts shall be of the screw, pressure, or solder type. When soldered connections are used, the conductors and terminal parts shall be mechanically secured before soldering. Only rosin based fluxes shall be used in soldering operations. Connections to screw type terminals shall be mechanically secured with a means to prevent loss of tightness.

3.4.4 Lubrication. All bearings (except sealed, permanently lubricated type), mating gears, and other moving parts shall be provided with means to ensure adequate lubrication. Re-circulating systems shall include filters that are cleanable or replaceable. Each lubricant reservoir shall have means for determining fluid levels. Oil holes, grease fittings, and filler caps shall be accessible. The manufacturer shall include and install all lubricants, fluids, greases, and other compounds in accordance with the best commercial practice of the producing industry (see 6.4). The lubrication systems shall be temporarily capped or stopped to prevent loss of lubricants during shipment. The manufacturer shall include a warning notice in the instruction manual and on the attachment to remove the temporary caps or stoppers before start-up or use of the attachment.

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3.4.5 Interchangeability. All parts shall be manufactured to definite standards, clearances, and tolerances in order that any such parts of a particular type or model may be replaced, interchanged, and adjusted without modification of the replacement parts or any other parts of the unit. When practical, all parts shall be permanently and legibly marked in accordance with MIL-STD-130.

3.4.6 Instruction plates. Instruction plates shall be permanently marked with bold-faced English letters on contrasting backgrounds and shall conform to MIL-STD-130.

3.5 Performance and product characteristics. The basic attachments, with appropriate standard and optional equipment, shall be capable of performing plain and universal milling, drilling, and boring operations through vertically adjustable three axis spindle positioning. The attachments shall be capable of standard keyseating or slotting with the required tooling. Feed for cutting operations shall be obtained from the co-joined machine except as otherwise specified herein.

3.5.1 Capacities. The capacities of the attachments shall not be less than the requirements specified in table I.

TABLE I. Attachment capacities.

Requirement	Size 1	Size 2
Adaptable for use on lathes of (inch) swing	10 to 13	13 to 18
Mounting surface to spindle center (inches)	1-13/16	2-5/8
Vertical adjustment (inches)	5	6
Spindle taper (Morse, ANSI B5.10)	2	3
Range of speed (rpm)	44 to 450	44 to 450
Optional speed range[1] (rpm)	540 to 5,500	540 to 5,500
Number of speeds (range)	13 (minimum)	13 (minimum)
Motors (see 3.6.4.1)	1/3 hp 115/230 volts 50/60 Hertz Single phase	3/4 hp 115/230 volts 50/60 Hertz Single phase
Optional motor[1]		3/4 hp 208/220/440 volts 50/60 Hertz Three phase

[1]Optional speed range and/or optional motor shall be furnished when specified (see 6.2).

3.5.2 Accuracy. The milling head shall be vertically adjustable and means shall be provided to lock the head at any point along the vertical travel without damaging the guideposts. The vertical adjustment shall work freely through the length of travel without sticking or binding. Locking shall be accomplished without deflecting the tool position more than 0.001 inch. The error in adjustment shall not exceed 0.0005 inch per inch of travel.

3.5.3 Milling with basic attachment. The basic attachment, when mounted on a suitable lathe, shall permit the following operations within the specified tolerances.

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3.5.3.1 Woodruff keyseat cutting. The basic attachment shall be capable of cutting Woodruff keyseats of the sizes specified below in 1020 steel round bar stock mounted between lathe centers in not more than five minutes for each cut. One keyseat shall be cut 1/4 inch from the end of the bar to the keyseat center. A second keyseat shall be cut at the midpoint of the bar. The cuts shall be free of chatter marks.

a. Size 1 - No. 204 Woodruff cutter, bar size 3/4 inch diameter by 6 inches long.
b. Size 2 - No. 406 Woodruff cutter, bar size 1 inch diameter by 8 inches long.

3.5.3.2 Keyway cutting. The basic attachment shall be capable of cutting keyways of the sizes specified below in one continuous operation in 1020 steel round bar stock, 30 inches long, mounted between centers.

a. Size 1 - 1/2 X 1/4 X 25 inches long at 1-3/4 inches per minute (ipm), bar diameter 2 inches.
b. Size 2 - 3/4 X 3/8 X 25 inches long at 1-1/2 ipm, bar diameter 3 inches.

3.6 Components. The attachments shall consist of the following principal components.

3.6.1 Base and column assembly. The base and column assembly shall consist of, but not be limited to, guideposts and a vertical adjusting screw. The base plate shall be adaptable to the lathe and shall provide rigidity for the guideposts. A vertically adjustable lead-screw shall be furnished, mounted on anti-friction thrust bearings with take-up for end play, and engaged by anti-backlash nuts in the sliding casting. The lead-screw shall be of hardened alloy 4140 steel or equivalent and shall have precision threads. The threads shall be uniform and have sufficient length of engagement to ensure the adjusted positional accuracy specified in 3.5.2. The lead-screw shall be turned by a hand crank and shall be provided with an adjustable collar graduated in increments of 0.001 inch that can be read continuously through the zero position.

3.6.2 Basic milling head. The basic milling head shall consist of a housing casting, spindle, gear drive, and motor support. The milling head shall be machined to house the spindle and gear drive assembly and to provide for motor attachment, guide posts slide, lead-screw, and positional clamps. The outer rim of the machined surface shall be graduated in increments of one degree for 30 degrees on each side of vertical zero. The spindle shall be hardened to Rockwell C45 to C52 and be precision ground, with the taper in the spindle nose conforming to ASME B5.10 and table I. The spindle and gear shafts shall be mounted on ball or roller bearings. All gears shall operate in an oil bath sealed to prevent entrance of foreign matter or leakage of oil when turned or rotated to any angle. The milling head shall provide for attachment of the drive motor directly or by motor mount. The main spindle shall provide for positive drive for all side mill, end mill, shell end mill, angle mill, and fly cutter arbors.

3.6.3 Belts and pulleys. When applicable, V-belt drives shall be furnished in accordance with RMA IP 25. Pulleys shall be of cast iron or aluminum alloy and shall be balanced. A belt guard shall be provided for protection of the operator at all positions during metalworking or grinding operations.

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3.6.4 Electrical equipment.

3.6.4.1 Motors. Motors shall conform to NEMA MG1, as applicable. The motors specified in table I shall be continuous duty induction squirrel-cage types. The operating voltage, frequency, and phase shall be as specified (see table I). Manual reset thermal overload protection shall be provided. A warning tag shall be attached to the power cord to identify the initial wiring of the motor (for example, 200 VAC, 60 Hz).

3.6.4.2 Starting switch. The starting switch shall be manual and have a rated capacity in accordance with NEMA ICS 1. If toggle switches are furnished, they shall be provided with a guard to prevent accidental engaging of the switch.

3.6.4.3 Electrical components. Electrical components of the attachments, including starters, relays, switches, wiring, and their enclosures, shall conform to and shall be located in accordance with NEMA ICS 1 and NFPA 79, as applicable.

3.6.4.4 Power supply cord. Each attachment shall be furnished with a power supply cord properly wired into the attachment. The cord shall have a voltage and current carrying capacity equal to, or more than that required by, the motors and NFPA 70. The power supply cord shall be at least 10 feet long and terminate with a three prong grounded plug.

3.6.4.5 Fungus control. Components of electrical circuitry shall be hermetically sealed and resistant to fungus growth and moisture.

3.7 Standard equipment. Each attachment shall be furnished with one each of the pieces of equipment specified in table II.

TABLE II. Standard equipment.

Component	Size 1[1]	Size 2[2]
Adapter, lathe compound slide, T-slot with 2 bolts	X	X
Adapter, lathe carriage cross-slide	X	X
Adapter for No. 1 Morse taper shank	X	
Adapter for No. 2 Morse taper shank		X
Arbor, side mill, with 1/8- and 3/8-inch spacing collars and nuts for 7/8-inch bore cutters	X	
Arbor, side mill, with 1/8- and 3/8-inch spacing collars and nuts for 1-inch bore cutters	X	
Arbor, side mill, with 1/8- and 3/8-inch spacing collars, standard keyway, and nuts for 7/8-inch bore cutters		X
Arbor, side mill, with 1/8- and 3/8-inch spacing collars, standard keyway, and nuts for inch bore cutters		X
Arbor, side mill, with 3-1/2 inch minimum mandrel, with 1/8-, 3/16-, 1/4-, 1/2-, 3/4-, and 1-1/4-inch spacing collars, and nuts for 1-inch bore cutters	X	X
Arbor, shell end mill, 1/2 inch	X	
Arbor, shell end mill, 3/4 inch		X

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TABLE II. Standard equipment - Continued.

Component	Size 1[1]	Size 2[2]
Arbor, for 3/8-inch diameter straight shanks, with setscrew	X	X
Arbor, for 1/2-inch diameter straight shanks, with setscrew	X	X
Arbor, flycutter, with 45° and 90° positions, sized to fit 1/4-inch minimum square cutting bit	X	
Arbor, flycutter, with 45° and 90° cutter positions, with 5/16-inch cutter bit		X
Arbor, with 0 to 1/2-inch capacity, with heavy duty ball bearing drill chuck	X	X
Belts, V drive (one extra set)	X	X
Spiral cut end mill, 5/8-inch diameter, with 1/2-inch diameter shank		X
Spiral cut end mill, 3/4-inch diameter, with 1/2-inch diameter shank		X
Pulleys, 3 each, two 4-step V-belt and one 3-step flat belt	X	X
T-bolts and nuts (for securing attachments to basic unit), 4 each	X	X

[1]Size 1 tool holders have a drawbar style #2 Morse taper shank.
[2]Size 2 tool holders have a drawbar style #3 Morse taper shank.

3.7.1 Cutters. Each attachment shall be furnished with one set of high speed cutters as specified in table III. In addition to those listed for size 1 units, one shell end mill (size 1-1/2 X 1-1/8 X 1/2 inch) shall be furnished. In addition to those listed for size 2 units, one shell end mill (size 2 X 1-3/8 X 3/4 inch) shall be furnished. Tolerance of cutters shall be in accordance with ASME B94.19.

TABLE III. Size of various high speed steel cutters.

Type of cutter	Cutter number	Nominal Diameter of cutter[1]	Diameter of cutter[1]	Width of face[1]	Diameter of shank[1]	Diameter of hole[1]	Length of cut[1]	Overall length[1]
Woodruff	204	1/2	-	1/16	1/2	-	-	2-1/16
Keatseat	305	5/8	-	3/32	1/2	-	-	2-3/32
Milling	406	3/4	-	1/8	1/2	-	-	2-1/8
Cutters	507	7/8	-	5/32	1/2	-	-	2-5/32
Plain metal slitting saw	-	-	4	1/8	-	1	-	-
Staggered tool side milling cutters	- - - -	- - - -	2-1/2 3 3 3	1/2 1/4 5/16 3/8	- - - -	7/8 1 1 1	- - - -	- - - -
Two flute single end mills for keyway cutting	- - - -	- - - -	1/8 1/4 3/8 1/2	- - - -	3/8 3/8 3/8 1/2	- - - -	3/8 1/2 9/16 1	2-5/16 2-5/16 2-5/16 3
60° dovetail cutter	-	-	1-1/4	-	1/2	-	-	-

[1]Dimensions are in inches.

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3.7.2 Wrench set. Each attachment shall be furnished with one set of wrenches that shall be capable of providing for the following functions:

a. Locking vertical adjustment of basic unit.
b. Locking belt tension adjustment.
c. Securing and removing V-belt sheaves and motor pulleys.
d. Securing and removing all arbors and arbor nuts.
e. Securing and releasing attachment fastening devices (see 3.8).
f. Securing and releasing belt-driven heads in shop cabinets.

3.8 Optional equipment. When specified (see 6.2), the following optional equipment shall be furnished. Equipment shall be of proper size and capacity to fit the size of basic head specified. The basic attachment and all accessories shall be capable of performing within the specified accuracy without damaging any of the parts. The method of securing equipment to the basic attachment shall be accomplished without disturbing adjustments such as spindle bearing mountings.

3.8.1 Ninety-degree universal milling head. The 90-degree universal milling head shall be capable of being securely mounted on the basic attachment. The head shall be adjustable for positioning the cutter at any point throughout 360 degrees and shall be self-indicating. The universal milling head shall permit the mounting of arbors and cutters specified herein and shall be capable of cutting the specified keyways in a 1020 steel round bar, 30 inches long, mounted between lathe centers. Cutting shall be at the specified range. The finish shall be not rougher than 125 microinches arithmetic average (aa). The cut shall extend full depth to the end of the bar with dimensions specified in table IV.

TABLE IV. Size of optional equipment.

Size 1						Size 2				
Cut dimensions						Cut dimensions				
Equipment	Width (inch)	Depth (inch)	Length (inch)	Feed rate (inch/min.)	Dia. (inch)	Width (inch)	Depth (inch)	Length (inch)	Feed rate (inch/min.)	Dia. (inch)
90° universal milling head	1/2	1/4	30	1	2	3/4	3/8	30	1	3
High speed end milling and drilling head	1/4	1/8	10	1	1	3/8	3/16	10	1-1/2	1-1/2
External grinding head	-	-	-	7	-	-	-	-	10	-
Internal grinder	-	3	-	-	1	-	4	-	-	1-1/4
Heavy-duty deep hole ginder	-	8	-	-	2-1/4	-	10	-	-	2-1/2
Internal keyseater and slotter (keyseat)	3/8	3/16	2-1/4	-	1-1/2	1/2	1/4	3	-	2
Feed table (keyway)	1/2	1/4	7	-	3/4 X 3/4	3/4	3/8	7	-	1 X 1

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3.8.2 High-speed end milling and drilling head. The high-speed milling and drilling head shall be capable of being securely mounted on the basic milling head, shall mount the same arbors and tools specified for the basic attachment, and shall permit cutter axis adjustment no less than 30 degrees above to 30 degrees below horizontal. The head shall provide for mounting and operating tools at either side of the attachment without removing bearings. The head shall provide a selection of spindle speeds arranged in approximate geometrical progression and of proper range to provide for drilling and end milling operations. The head shall be capable of cutting the specified keyways at the specified rate in a 1020 steel round bar, 15 inches long, mounted between lathe centers (see 3.5.3.2 for minimum bar diameters). One keyway shall extend full depth to the end of the bar and the other keyway shall terminate 1/4 inch from the end of the bar. The cut surfaces shall not be rougher than 125 microinches aa with dimensions specified in table IV.

3.8.3 External grinding head. The external grinding head shall be a precision machine capable of cylindrical, tool, and thread grinding. The head shall be capable of being mounted on the basic milling head, shall permit grinder axis adjustment not less than 30 degrees above to 30 degrees below horizontal, and shall be driven by the basic milling head motor. Provision shall be made to permit grinding from either side of the attachment without disturbing the spindle-bearing mounting. A minimum of six spindle speeds shall be provided. Grit, bond, diameter, and the speed of the grinding wheels furnished shall be suitable for efficient grinding at the full diameter of the wheels and after the wheels have been dressed to 2/3 of their original diameter. Grinding wheel speeds shall not exceed the requirements of ANSI B7.1. The following grinding wheels shall be furnished with each head:

a. Two straight wheels.
b. One flare cup wheel.
c. One dish wheel.

A grinding wheel guard conforming to the requirements of ANSI B7.1 shall be provided. The grinder shall be capable of removing not less than 0.020 inch from the diameter of a 1-inch (+1/4 inch, -0 inch) shaft of 1045 steel that has been heat treated to Rockwell C40 to C50. Length of ground surface shall be not less than 10 inches. The specified diameter reduction shall be accomplished in not more than 30 minutes and the resulting surface finish shall not exceed 32 microinches aa. Feed rate shall be as specified in table IV.

3.8.4 Internal grinding head. The internal grinding head shall be similar in construction to the external grinding head and subject to the same requirements, except that the spindle shall be extended for grinding bores of the diameter and depth specified by table IV for 1045 steel that has been heat treated to Rockwell C40 to C50. The grinder shall be capable of removing 0.020 inch from the bore diameter in not more than 30 minutes. Surface finish shall not be rougher than 16 microinches aa. A minimum of six spindle speeds shall be provided. Two straight wheels shall be furnished with each grinder.

3.8.5 Heavy-duty deep-hole grinder. The heavy-duty deep-hole grinder shall be capable of being mounted on the basic milling head and shall conform in general characteristics of design to other grinding heads specified herein. The grinder shall be capable of grinding bores of the diameter and depth specified by table IV for 1045 steel that has been heat treated to Rockwell C40 to C50. The grinder shall be capable of removing 0.020 inch from the bore diameter in not

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more than 30 minutes. Surface finish shall not be rougher than 16 microinches aa. A minimum of six spindle speeds shall be provided. A grinding wheel shall be furnished that will allow this requirement to be accomplished.

3.8.6 Dividing head. The dividing head shall be adaptable to fit 1-3/8 inch diameter lathe spindle bores. The head shall be furnished with one machined adapter for a 1-3/8 inch bore spindle and three unfinished adapters to cover a range of 1-1/8 inches to 3-1/2 inches. The dividing head shall be capable of indexing all divisions from 1 to 50, inclusive, and divisions 55 to 120 by multiples of five. The dividing head shall be indexed by degrees (360 equal divisions), 1/2 degree (720 equal divisions), and 1/3 degree (1,080 equal divisions). Positioning error shall be not greater than 1/2 minute of one degree (30 seconds).

3.8.7 Internal keyseater and slotter. The internal keyseater shall be capable of cutting a standard internal keyway as listed in table V in the bore size for which it is standard, for a length up to 1-1/2 times the bore diameter as listed in ASME B17.1. Sizes 1 and 2 shall have a stroke adjustable from 0 to 4 inches. The keyseater and slotter shall be capable of being securely mounted on the basic attachment and of being locked in position to provide strokes of any angle in relation to the base mounting surface. The necessary cutters and tool bits shall be included to cut each of the internal keyways specified below in table V for both sizes 1 and 2. The keyseater and slotter shall be capable of cutting the keyseats specified in table IV in the specified bores of 1020 steel in not more than five minutes. The surface finish shall not exceed 125 microinches aa.

TABLE V. Internal keyway dimensions.

Dimensions	Size 1	Size 2
1/8 X 1/16 inch	X	X
3/16 X 3/32 inch	X	X
1/4 X 1/8 inch	X	X
5/16 X 5/32 inch	X	X
3/8 X 3/16 inch	X	X
1/2 X 1/4 inch	X	X

3.8.8 Feed table. The feed table shall be capable of being mounted to the basic attachment to create a portable machining unit. The feed table shall consist of a base, a lower carriage that shall travel longitudinally in relation to the base, an upper carriage that shall travel across the base, and the necessary screws, screw blocks, anti-backlash nuts, and gibs to equip it to meet the accuracy and performance specified herein. The base shall be capable of being rigidly bolted or clamped to a supporting surface without interfering with the movement of either carriage. The lower carriage shall provide travel of not less than 3 inches at 90 degrees to the direction of travel of the lower carriage. A means of locking at any point in either direction of travel shall be provided. Locking either carriage shall not deflect the basic attachment more than 0.001 inch. Feed shall be provided with a graduated collar or dial similar in form and marking to that specified for vertical travel of the basic unit. Each collar shall be capable of being set at 0 at any point of the carriage travel. Travel, as indicated by either collar, shall be accurate within 0.005 inch per inch of travel. The basic head, when mounted on the feed table, shall permit cutting the

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keyways specified in table IV in a 1020 steel square bar in not more than 5 minutes for size 1 or 6 minutes for size 2. Feed shall be manually accomplished. The total height from the mounting surface to the top surface shall not exceed 3-3/8 inches.

3.8.9 Grinding gage. A grinding gage (tooth stop rest) for milling cutters shall be universally adjustable and sufficiently rigid to provide accuracy in grinding cutters.

3.8.10 Diamond dresser. The diamond wheel dresser shall consist of a frame that can be clamped in a suitable position for dressing grinding wheels specified herein and a properly mounted industrial diamond. The diamond shall be free from faults that may affect the serviceability of the dresser. Diamonds shall be natural octahedron with not less than six cutting edges. Diamonds shall be at least 1/4 carat if the maximum size of the wheels furnished is 4 inches or less in diameter, at least 3/10 carat if the maximum size wheels furnished is 6 inches, and not less than 1/2 carat for 7 inch wheels.

3.8.11 Shop cabinet. A shop cabinet shall be provided of sufficient size, with brackets, clamps, and any other holding devices required to store the basic attachment and all specified and optional equipment, spare parts, and wrenches in an orderly manner. The cabinet shall be constructed so that the various stored parts will not be displaced by rough handling due to shipping. The cabinet shall be designed to provide easy access to its contents. It shall be adequate to protect all contents from air-blown metal chips, coolant, dirt, grease, and other contaminants encountered during machine shop operations.

3.9 Painting and finishing. Before painting, all surfaces shall be clean and be free of all foreign matter detrimental to painting. Unless otherwise specified (see 6.2), at least one coat of primer (or a phosphate or chromate base) and one coat of enamel or equivalent (for example, epoxy enamel) shall be applied.

3.10 Product identification. The attachments shall be marked for product identification in accordance with MIL-STD-130 and unless otherwise specified (see 6.2), shall include the national stock number.

3.11 Workmanship. Workmanship shall be of the quality necessary to produce complete attachments of good appearance with safe and efficient operating characteristics that are customarily found in high grade commercial items of a similar nature.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the acquisition order (see 6.2), the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the acquisition order (see 6.2), the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the government. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to the prescribed requirements.

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4.1.1 Responsibility for compliance. All items must meet all requirements of section 3. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the government for acceptance comply with all requirements of the contract. Sampling in the quality conformance does not authorize submission of known defective materiel, either indicated or actual, nor does it commit the government to acceptance of defective materiel.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 3.1), the number of units to be provided for inspection shall be specified (see 6.2). The attachments provided for first article inspection shall undergo examinations for defects specified in table VI, and the tests specified in table VII. Attachments displaying any defect shall be rejected.

TABLE VI. Defects.

Defect	Requirement
Gears	3.2.4
Construction	3.4
Castings and forgings	3.4.1
Electrical connections	3.4.3
Lubrication	3.4.4
Accuracy	3.5.2
Base and column assembly	3.6.1
Basic milling head	3.6.2
Motors	3.6.4.1
Product identification	3.10
Workmanship	3.11

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TABLE VII. Test requirements.

Test	Test paragraph
Motor	4.6.1.1
Milling	4.6.1.2
Woodruff keyseat cutting with basic attachment	3.5.3.1 and 4.6.1.2.1
Keyway cutting with basic attachment	3.5.3.2 and 4.6.1.2.2
Keyway cutting with 90 degree universal milling head	3.8.1 and 4.6.1.2.3
Keyway cutting with the high speed end milling and drilling head	3.8.2 and 4.6.1.2.4
Performance test of external grinder	3.8.3 and 4.6.1.3.1
Performance test of internal grinder	3.8.4 and 4.6.1.3.2
Performance test of deep-hole grinder	3.8.5 and 4.6.1.3.3
Performance of dividing head	3.8.6 and 4.6.1.4
Performance test of internal keyseater and slotter	3.8.7 and 4.6.1.5
Performance of feed table	3.8.8 and 4.6.1.6

4.4 Quality conformance inspection. Unless otherwise specified (see 6.2), quality conformance inspection shall be applied to production items offered for acceptance under the contract and shall consist of (a) through (d) below. A sample of the attachments shall be selected in accordance with MIL-STD-1916 inspection level I. Failure of any attachment to pass an examination or test shall be cause for rejection of the item.

a. Product examination (see 4.5).
b. Motor tests (see 4.6.1.1).
c. Milling test (see 4.6.1.2).
d. Woodruff keyseat cutting test with basic attachment (see 4.6.1.2.1).

4.5 Product examination. Visually, dimensionally, and manually examine each attachment to determine conformance with the requirements of 3.2 through 3.5.2 and 3.6 through 3.11. Visual examination shall include verification of completeness of manufacture and assembly, conformance to the specified standards, adequacy of markings, proper cleaning, and freedom from the identified defects. Dimensional examination includes dimensions as specified. Manual examination shall include the operation of movable parts by hand to ensure proper functioning. The examination provisions may be applied at the earliest practical point in manufacture at which it is feasible to inspect for acceptance without risk of change in the characteristic by subsequent operations. Failure of the contractor to provide objective evidence that the attachment and its components have passed the examinations prescribed for them by the contractor's inspection system shall be cause for rejection. A certificate of conformance may be accepted in lieu of the visual or manual requirements of 3.2.4, 3.2.5.1, 3.4.1, 3.4.3, and 3.4.5. In addition, failure of the contractor to provide objective evidence that all parts are manufactured to definite standards, clearances, and tolerances so that no replacement part will degrade the form, fit, or function of the end item (see 3.4.5) shall be cause for rejection.

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4.5.1 Evidence of compliance with OSHA standards. Evidence of compliance with applicable safety and health requirements shall be reviewed (see 3.2.5.1).

4.6 Test methods.

4.6.1 Performance tests. The attachment and the standard and optional equipment (see 3.7 and 3.8) shall be tested as follows for compliance with the requirements of 3.5.

4.6.1.1 Motor tests. The contractor shall certify that all motors have been tested and comply with the applicable requirements of NEMA MG 1 (see 3.6.4.1).

4.6.1.2 Milling tests. The basic attachment shall be set up on a suitable lathe (see 3.5.3). Keyways of specified size and lengths shall be cut in one continuous operation at the specified rate in 1020 steel of a diameter appropriate for use with a plain parallel key in accordance with ANSI B17.1, except as specified herein. Cuts shall be straight and true and the finish shall not exceed 125 microinches aa. A follower rest or other method of bracing the stock may be used. Setup and milling shall be in accordance with information contained in the manufacturer's instructions.

4.6.1.2.1 Woodruff keyseat cutting with basic attachment. Two Woodruff keyseats of appropriate size shall be cut to determine compliance with 3.5.3.1.

4.6.1.2.2 Keyway cutting with basic attachment. A keyway of appropriate size shall be cut to determine compliance with 3.5.3.2.

4.6.1.2.3 Keyway cutting with 90 degree universal milling head. The universal milling head shall make the specified cuts with the spindle in a vertical position to determine compliance with 3.8.1.

4.6.1.2.4 Keyway cutting with the high-speed end milling and drilling head. The end milling head shall make two cuts to determine compliance with 3.8.2.

4.6.1.3 Grinding tests. The basic attachment, with appropriate grinding head, shall be set up on a suitable lathe. Setup and method of obtaining power feed rates shall be in accordance with information contained in the manufacturer's instructions.

4.6.1.3.1 Performance test of external grinder. The work piece shall be 15 inches long and rough turned to present a test surface not less than 10 inches long. The test surface diameter shall be reduced 0.020 inch to determine compliance with 3.8.3.

4.6.1.3.2 Performance test of internal grinder. The internal grinder shall be tested to determine compliance with 3.8.4.

4.6.1.3.3 Performance test of deep-hole grinder. The deep-hole grinder shall be tested to determine compliance with 3.8.5.

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4.6.1.4 Performance of dividing head. The dividing head shall be properly mounted on a lathe and test cuts shall be made in the periphery of a 6-inch diameter work piece held by the lathe chuck. The cuts shall be made with a side-milling cutter and shall be 1/4 inch wide and 1/8 inch deep. Cuts shall be spaced at 120 degrees, 90 degrees, 60 degrees, 40 degrees, and 20 degrees from the first cut. The total error in location of each cut shall not be more than one-half minute of one degree when measured from the base cut (see 3.8.6).

4.6.1.5 Performance test of internal keyseater and slotter. The keyseater and slotter shall be tested to determine compliance with 3.8.7.

4.6.1.6 Performance test of feed table. The basic attachment, with feed table, shall be securely attached to a suitable surface fitted with a vise or other means for positioning and holding the appropriate work piece. A keyway shall be cut to determine compliance with 3.8.8.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the acquisition order (see 6.2).

6. NOTES

INFORMATION FOR GUIDANCE ONLY. (This section contains information of a general or explanatory nature that is helpful, but is not mandatory.)

6.1 Intended use. Attachments covered by this standard are intended for use in milling, drilling, boring, gear cutting, thread milling, keyseating, slotting, and external and internal grinding operations.

6.2 Acquisition requirements. The acquisition order should specify the following:

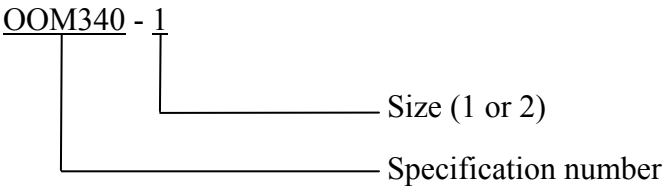
a. Title, number, revision, and date of this standard.
b. Size required (see 1.2).
c. First article, when required (see 3.1).
d. US and SI measuring and indicating device graduations required (see 3.2.2).
e. Optional speed range and motors, when required (see 3.5.1 and table I).
f. Optional equipment required (see 3.8).
g. Painting and finishing, if different (see 3.9).
h. Product identification, if different (see 3.10).
i. Inspection responsibility, if other than contractor (see 4.1).
j. Inspection facilities, if other than contractor (see 4.1).
k. Number of units to be provided for first article inspection (see 4.3).
l. Quality conformance inspection, if different (see 4.4).
m. Packaging (see 5.1).

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6.3 Repair part manuals. Repair part identification in repair part manuals should include the prime contractor's name, nomenclature, part number, and price. Repair part identification should also include the name of the original equipment/part manufacturer (OEM), federal supply code for manufacturers (FSCM), nomenclature, part number, and price of the part immediately after the prime contractor's part identification.

6.4 Standard lubricants. The attachment manufacturer should specify and use to the maximum extent possible those lubricants (see 3.4.4) that are presently in the Department of Defense supply system. MIL-HDBK-113, "Guide for the Selection of Lubricants, Functional Fluids, Preservatives, and Speciality Products for Use in Ground Equipment Systems", MIL-HDBK-267, "Guide for Selection of Lubricants and Hydraulic Fluids for Use in Shipboard Equipment", and MIL-HDBK-838, "Lubrication of Military Equipment" may be used as selection guides.

6.5 Part or identification number. The following PIN procedure is for government purposes and does not constitute a requirement for the contractor.



6.5 Subject term (key word) listing.

boring
feed table
gear cutting
keyseating
lathe
thread

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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Custodians: Army - AR Navy - SH Air Force - 99	GSA - 6FEE
Review Activities: Army - CE, CR4 Navy - MC, Air Force - 84	Preparing Activity: DLA - GS6 (Project 3460-0212)

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